A2 · Unit 5 · WPH15 Thermodynamics, Radiation, Oscillations & Cosmology

Gravitational Fields

Every mass in the universe pulls on every other mass. That pull keeps your feet on the floor, the Moon going round the Earth and the Earth going round the Sun. In this topic we describe gravity as a **field**, work out how strong it is at any distance from a planet, meet **gravitational potential**, and then use Newton's laws to explain how satellites, moons and planets orbit. Electric fields follow almost the same rules, so we finish by comparing the two.

Gravitational fields and field strength

Nothing has to touch you for the Earth to pull on you: a skydiver speeds up with nothing attached to them. Physicists describe this "action at a distance" with the idea of a **field**. Every mass fills the space around it with a gravitational field, and any other mass placed in that field feels a force.

DEFINITION

Gravitational field and field strength

A **gravitational field** is a region in which a mass experiences a force.

The **gravitational field strength** g at a point is the gravitational force per unit mass acting on a small test mass placed at that point.

Gravitational field strength

FORMULA SHEET

$$g = \frac{F}{m}$$

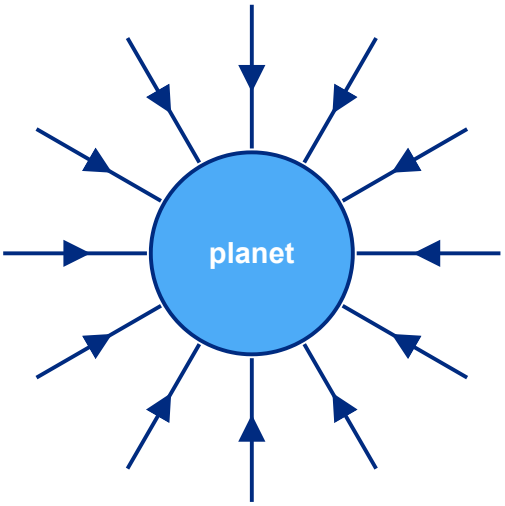
g: gravitational field strength / N kg^{-1} F: gravitational force on the mass / N m: mass placed in the field / kg

Field strength is a **vector**. It points the same way as the force on a mass, which is always **towards** the mass that creates the field. We imagine a **small** test mass so that its own field doesn't disturb the field we are trying to measure.

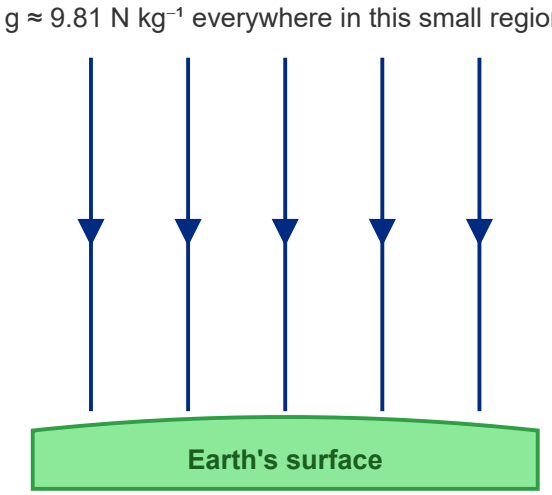
KEY IDEA

Field strength is the acceleration of free fall

The units N kg^{-1} and m s^{-2} are the same thing (1 N = 1 kg m s^{-2}). Newton's second law says $a = F/m$, which is exactly $g = F/m$. So "the field strength at the Earth's surface is 9.81 N kg^{-1}" and "objects fall with an acceleration of 9.81 m s^{-2}" say the same thing. Every object at the same place gets the same acceleration, whatever its mass, because doubling m doubles the force too.



radial field: lines spread out, field gets weaker



near the surface: almost uniform

Field lines show the direction of the force on a mass, and their spacing shows the strength. Around a planet the field is **radial**: the lines point in towards the centre and spread out with distance. Over a small region near the surface the lines are almost parallel and equally spaced, so the field is nearly **uniform**. That is why at AS you could use $g = 9.81\ \text{N kg}^{-1}$ and $\Delta E = mg\Delta h$.

EXAM TIP

When you draw gravitational field lines, always put arrows on them pointing **towards** the mass. Lines never cross, and they meet the surface of a spherical planet at right angles, heading for its centre.

Newton's law of universal gravitation

Newton's big idea (1687) was that the force that makes an apple fall is the same force that holds the Moon in its orbit. His law says that any two point masses attract each other with a force that is

- **proportional to the product of their masses**: double either mass and the force doubles;
- **inversely proportional to the square of the distance between their centres**: double the distance and the force drops to a quarter.

Newton's law of universal gravitation

FORMULA SHEET

$$F = \frac{Gm_1m_2}{r^2}$$

F: force on each mass / N m_1, m_2: the two masses / kg r: distance between their centres / m

$G = 6.67 \times 10^{-11}\ \text{N m}^2\,\text{kg}^{-2}$

DEFINITION

Newton's law of gravitation

Two point masses attract each other with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

KEY IDEA

Why the *square* of the distance?

Picture the field lines leaving a planet. At a distance r they are spread over the surface of a sphere of area $4\pi r^2$. Double the distance and the same number of lines is shared over four times the area, so the field, and the force on a mass, is a quarter as strong. Treble it and you get one ninth. Anything that spreads out evenly in three dimensions without being absorbed follows an inverse-square law. You will meet the same $4\pi r^2$ in electric fields and in the intensity of light from stars.

G is tiny, which is why you never notice the pull between everyday objects. Two 70 kg people standing 1 m apart attract each other with about 3×10^{-7} N, roughly the weight of a grain of sand. Gravity only matters when at least one of the masses is enormous, like a planet or a star.

The force obeys **Newton's third law**: the Earth pulls the Moon, and the Moon pulls the Earth with a force of exactly the same size in the opposite direction. The effects are very different only because the masses are.

COMMON MISTAKE

r is measured between the **centres** of the masses. For a satellite 400 km above the Earth, $r = 6.37 \times 10^6 + 4.0 \times 10^5 = 6.77 \times 10^6$ m, not 4.0×10^5 m. And don't forget to square it.

WORKED EXAMPLE

The Earth (mass 5.97×10^{24} kg) and the Moon (mass 7.35×10^{22} kg) are 3.84×10^8 m apart. Calculate the gravitational force between them and the acceleration each one has because of it.

Show solution

$$F = \frac{Gm_1m_2}{r^2} = \frac{6.67 \times 10^{-11} \times 5.97 \times 10^{24} \times 7.35 \times 10^{22}}{(3.84 \times 10^8)^2} = 1.98 \times 10^{20} \text{ N}$$

The same size of force acts on each body (Newton's third law), so use $a = F/m$ for each:

$$a_{\text{Moon}} = \frac{1.98 \times 10^{20}}{7.35 \times 10^{22}} = 2.70 \times 10^{-3} \text{ m s}^{-2}, \qquad a_{\text{Earth}} = \frac{1.98 \times 10^{20}}{5.97 \times 10^{24}} = 3.32 \times 10^{-5} \text{ m s}^{-2}$$

The Moon's acceleration is the centripetal acceleration that keeps it in orbit. The Earth's is about 80 times smaller: it makes the Earth wobble slightly as the Moon goes round.

The field around a point mass: g = Gm/r²

Now combine the two equations you have met so far. This is a derivation the specification says you must be able to do, and it only takes three lines.

Deriving g = Gm/r² for a point mass

SPEC: YOU MUST BE ABLE TO DERIVE THIS

1. Place a small test mass m a distance r from a point mass M. Newton's law of gravitation gives the force on it:

$$F = \frac{GMm}{r^2}$$

2. Gravitational field strength is defined as force per unit mass, so divide by m:

$$g = \frac{F}{m} = \frac{GMm}{r^2} \cdot \frac{1}{m}$$

3. The test mass cancels:

$$\boxed{g = \frac{GM}{r^2}}$$

The field depends only on the mass that creates it and how far away you are, not on what you put there. That is why a hammer and a feather dropped together on the Moon land together.

4. The field points towards M. Outside a spherical planet or star, the field is the same as if all its mass were concentrated at its centre, so the equation works for any r from the surface outwards, with r measured from the centre.

Newton proved this "shell theorem" with some clever geometry. The proof is not in the spec; you just need to know you can treat a sphere as a point mass.

Radial gravitational field

FORMULA SHEET

$$g = \frac{Gm}{r^2}$$

g: field strength / N kg⁻¹ m: mass creating the field / kg r: distance from its centre / m

Because $g \propto 1/r^2$, going from the surface ($r = R$) to $r = 2R$ divides g by 4, and $r = 3R$ divides it by 9. One Earth radius above the ground, g is only about 2.45 N kg⁻¹.

WORKED EXAMPLE

The Earth's radius is 6.37×10^6 m and g at its surface is 9.81 N kg⁻¹. (a) Calculate the mass of the Earth. (b) The International Space Station orbits 408 km above the surface. Calculate g at that height.

Show solution

(a) Rearrange $g = GM/R^2$:

$$M = \frac{gR^2}{G} = \frac{9.81 \times (6.37 \times 10^6)^2}{6.67 \times 10^{-11}} = 5.97 \times 10^{24}\ \text{kg}$$

(b) Measure from the centre: $r = 6.37 \times 10^6 + 0.408 \times 10^6 = 6.778 \times 10^6$ m.

$$g = \frac{GM}{r^2} = \frac{6.67 \times 10^{-11} \times 5.97 \times 10^{24}}{(6.778 \times 10^6)^2} = 8.67\ \text{N kg}^{-1}$$

That is still 88% of the value on the ground. Gravity has certainly not "switched off" up there, which matters when we explain why astronauts float.

Two masses: where do the fields cancel?

Field strengths are vectors, so where two fields overlap you add them as vectors. On the line between the Earth and the Moon the two pulls point in opposite directions. Close to the Earth its pull wins easily, but somewhere near the Moon the two are equal and opposite and the resultant field is zero. This is the **neutral point**.

WORKED EXAMPLE

Using the data from the first worked example, find how far from the centre of the Earth the resultant gravitational field on the Earth–Moon line is zero.

Show solution

Let the point be a distance x from the Earth, so $d - x$ from the Moon, with $d = 3.84 \times 10^8$ m. The two field strengths are equal there:

$$\frac{GM_E}{x^2} = \frac{GM_M}{(d-x)^2} \quad \Rightarrow \quad \frac{d-x}{x} = \sqrt{\frac{M_M}{M_E}} = \sqrt{\frac{7.35 \times 10^{22}}{5.97 \times 10^{24}}} = 0.111$$

Taking the square root of both sides keeps the algebra linear. Use the positive root: the point is between the two bodies.

$$d = 1.111\,x \quad \Rightarrow \quad x = \frac{3.84 \times 10^8}{1.111} = 3.46 \times 10^8\ \text{m}$$

That is 90% of the way to the Moon, about 3.8×10^7 m from the Moon's centre. G cancelled, so you didn't need its value.

Gravitational potential

Lift a book and you do work against gravity; let go and the field does work on it as it falls. **Gravitational potential** tells you how much work per kilogram is involved in getting to each point in a field. It works like height on a map: the field pulls masses "downhill" towards lower potential.

DEFINITION

Gravitational potential

The gravitational potential at a point is the work done per unit mass in bringing a small test mass from infinity to that point. It is a **scalar**, measured in J kg⁻¹.

Gravitational potential in a radial field

FORMULA SHEET

$$V_{\text{grav}} = -\frac{Gm}{r}$$

V_{grav}: potential / J kg⁻¹ m: mass creating the field / kg r: distance from its centre / m

KEY IDEA

Why is the potential negative?

We choose the potential to be **zero at infinity**, far from every mass. Gravity is attractive, so as a mass falls in from infinity the field does the work; to bring it in slowly you would have to hold it back, which means *you* do negative work. So at every finite distance $V < 0$, and the closer you get to the mass the more negative it becomes. A mass near a planet sits at the bottom of a "potential well", and you must supply energy to lift it out.

Notice that V goes as $1/r$, not $1/r^2$: the potential fades away more slowly than the field strength. At the Earth's surface $V = -6.25 \times 10^7$ J kg^{-1} and at twice that distance it is half as big, while g has dropped to a quarter.

Potential difference and potential energy

To move a mass m between two points you must do work equal to the mass times the change in potential: $\Delta W = m\,\Delta V$. It follows that a mass m at a point where the potential is V_{grav} has gravitational potential energy

$$E_{\text{grav}} = mV_{\text{grav}} = -\frac{GMm}{r}$$

This is also negative, and zero at infinity. Close to the Earth, over a height change that is small compared with the radius, $\Delta V \approx g\,\Delta h$, which gives back the AS result $\Delta E = mg\Delta h$. Over large distances g changes, so you must use ΔV instead.

WORKED EXAMPLE

A 1000 kg satellite is to be placed in a circular orbit 400 km above the Earth (radius 6.37×10^6 m, mass 5.97×10^{24} kg). (a) Calculate the increase in its gravitational potential energy. (b) Compare your answer with $mg\Delta h$. (c) Calculate the kinetic energy it needs in orbit.

Show solution

(a) $GM = 6.67 \times 10^{-11} \times 5.97 \times 10^{24} = 3.982 \times 10^{14}$ N m^2 kg^{-1}. Find the potential at each radius:

$$V_1 = -\frac{3.982 \times 10^{14}}{6.37 \times 10^6} = -6.251 \times 10^7 \text{ J kg}^{-1}, \qquad V_2 = -\frac{3.982 \times 10^{14}}{6.77 \times 10^6} = -5.882 \times 10^7 \text{ J kg}^{-1}$$

$$\Delta E = m\,\Delta V = 1000 \times (-5.882 \times 10^7 - (-6.251 \times 10^7)) = 3.69 \times 10^9 \text{ J}$$

(b) $mg\Delta h = 1000 \times 9.81 \times 4.0 \times 10^5 = 3.92 \times 10^9$ J. This is about 6% too big, because g gets smaller as the satellite rises (it is 8.69 N kg⁻¹ at 400 km).

(c) In a circular orbit $v^2 = GM/r$ (see the orbits section below), so

$$E_k = \tfrac{1}{2}mv^2 = \frac{GMm}{2r} = \frac{3.98 \times 10^{14} \times 1000}{2 \times 6.77 \times 10^6} = 2.94 \times 10^{10} \text{ J}$$

Eight times the potential energy. Getting up to orbit is the easy part; getting fast enough to stay there is what needs the huge rockets.

Equipotentials and field lines

An **equipotential** is a surface (a line on a flat diagram) on which the potential is the same everywhere. Around a single planet they are spheres. Three things to know:

- No work is done moving a mass along an equipotential, because $\Delta V = 0$.
- Equipotentials are always **perpendicular to field lines**. If they weren't, the field would have a component along the equipotential and would do work there.
- If you draw them at equal steps of ΔV, they crowd together where the field is strong, like contour lines on a steep hill.

Field strength is the potential gradient

The field strength tells you how steeply the potential changes with distance:

$$g = -\frac{\Delta V}{\Delta r}$$

On a graph of V against r, the gradient at any point equals the size of g there. The minus sign says the field points the way the potential *decreases*, which is towards the mass, just as a ball rolls downhill. You can check the units: J kg⁻¹ m⁻¹ = N m kg⁻¹ m⁻¹ = N kg⁻¹. The same link between field and potential holds for electric fields.

Orbital motion

Newton imagined a cannon on top of a very high mountain, firing balls horizontally. A slow ball curves down and hits the ground nearby. A faster one lands further away. Fire it fast enough and, as it falls, the curved Earth drops away beneath it at the same rate: it keeps falling but never lands. That is an orbit. **An orbiting satellite is in continuous free fall.**

KEY IDEA

Newton's three laws in an orbit

- **First law:** without a force the satellite would fly off in a straight line along the tangent. Something must keep pulling it round.
- **Second law:** that something is gravity, the only force acting. It is the resultant force, so it causes an acceleration towards the centre: it provides the **centripetal force** mv^2/r (see Circular Motion).

- **Third law:** the satellite pulls the Earth with an equal and opposite force. Because the Earth is so massive, its acceleration is negligible.

Orbital speed of a satellite

1. For a satellite of mass m in a circular orbit of radius r round a planet of mass M, gravity provides the centripetal force:

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

2. Cancel m and one factor of r:

$$v^2 = \frac{GM}{r}$$

3. Take the square root:

$$\boxed{v = \sqrt{\frac{GM}{r}}}$$

The satellite's mass has cancelled, so a spanner and a space station in the same orbit move at the same speed. A bigger r means a smaller v: higher orbits are slower.

EXAM TIP

Why do astronauts float?

Not because there is no gravity: g at the ISS is 8.7 N kg^{-1}. The astronauts and the station are falling together with the same acceleration, so the floor doesn't need to push on them, and they feel weightless. Examiners like the phrase "in free fall, with the same acceleration as the spacecraft".

Period and radius: T² ∝ r³

A satellite goes once round its orbit, a distance $2\pi r$, in one period T, so $v = 2\pi r/T$. Put that into the orbital speed equation and you get a relationship between the period and the radius of any orbit.

Period of a circular orbit

1. Start from $v^2 = GM/r$ and substitute $v = 2\pi r/T$:

$$\frac{4\pi^2 r^2}{T^2} = \frac{GM}{r}$$

2 Rearrange to make T^2 the subject:

$$\boxed{T^2 = \frac{4\pi^2}{GM} r^3}$$

Kepler discovered $T^2 \propto r^3$ from planetary observations in 1619 ("Kepler's third law"), almost 70 years before Newton's law of gravitation explained it.

3 So for everything orbiting the same central body, a graph of T^2 against r^3 is a straight line through the origin with

$$\text{gradient} = \frac{4\pi^2}{GM} \quad \Rightarrow \quad M = \frac{4\pi^2}{G \times \text{gradient}}$$

This is how astronomers "weigh" planets and stars: time how long a moon takes to go round and measure the size of its orbit. You don't need to know anything about the moon itself.

WORKED EXAMPLE

Io orbits Jupiter with a period of 1.77 days at a radius of 4.22×10^8 m. Calculate the mass of Jupiter.

Show solution

Convert the period to seconds: $T = 1.77 \times 24 \times 3600 = 1.529 \times 10^5$ s. Rearrange $T^2 = 4\pi^2 r^3 / GM$:

$$M = \frac{4\pi^2 r^3}{GT^2} = \frac{4\pi^2 \times (4.22 \times 10^8)^3}{6.67 \times 10^{-11} \times (1.529 \times 10^5)^2} = 1.90 \times 10^{27} \text{ kg}$$

About 320 times the mass of the Earth.

Geostationary orbits

A **geostationary** satellite stays above the same point on the Earth's surface. For that it must

- have a period of **24 hours**, the same as the Earth's rotation;
- orbit above the **equator**;
- move in the **same direction** as the Earth rotates (west to east).

Because T is fixed, $T^2 \propto r^3$ fixes the radius too: there is only one geostationary orbit. Communications and TV satellites use it, so a dish on your roof can point at one spot in the sky and never move. Low orbits like the ISS (about 90 minutes per orbit) are better for photographing the ground in detail, and GPS satellites take about 12 hours.

WORKED EXAMPLE

Calculate the height above the Earth's surface of a geostationary orbit, and the speed of a satellite in it. Mass of Earth $= 5.97 \times 10^{24}$ kg, radius of Earth $= 6.37 \times 10^6$ m.

Show solution

$T = 24 \times 3600 = 86\,400$ s. Rearrange $T^2 = 4\pi^2 r^3 / GM$ for r:

$$r^3 = \frac{GMT^2}{4\pi^2} = \frac{6.67 \times 10^{-11} \times 5.97 \times 10^{24} \times (86\,400)^2}{4\pi^2} = 7.53 \times 10^{22} \text{ m}^3$$

$$r = \sqrt[3]{7.53 \times 10^{22}} = 4.22 \times 10^7 \text{ m}$$

This is measured from the centre, so subtract the Earth's radius:

$$h = 4.22 \times 10^7 - 6.37 \times 10^6 = 3.59 \times 10^7 \text{ m} \quad (\text{about } 36\,000 \text{ km})$$

$$v = \frac{2\pi r}{T} = \frac{2\pi \times 4.22 \times 10^7}{86\,400} = 3.07 \times 10^3 \text{ m s}^{-1}$$

COMMON MISTAKE

Two classic slips in orbit questions: leaving T in hours or days (it must be in **seconds**), and giving r as the answer when the question asks for the **height above the surface**.

Comparing gravitational and electric fields

Gravitational fields act on **mass**; electric fields act on **charge**. Apart from that, the maths is almost identical, so learning one helps you with the other (see Electric Fields).

	Gravitational field	Electric field
Acts on	mass	charge
Field strength	$g = F/m$, N kg⁻¹	$E = F/Q$, N C⁻¹
Force between two point objects	$F = \frac{Gm_1m_2}{r^2}$	$F = \frac{Q_1Q_2}{4\pi\varepsilon_0 r^2}$
Field of a point object	$g = \frac{Gm}{r^2}$	$E = \frac{Q}{4\pi\varepsilon_0 r^2}$
Potential in a radial field	$V = -\frac{Gm}{r}$: always negative	$V = \frac{Q}{4\pi\varepsilon_0 r}$: positive or negative
Attract or repel?	always attracts	like charges repel, unlike attract
Constant	$G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}	$\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^9$ N m^2 C^{-2}

Uniform field	near a planet's surface (approximately)	between parallel plates, $E = V/d$

KEY IDEA

Similarities and differences

Similar: both follow an inverse-square law for a point source; both radial fields look the same on a diagram; both have equipotentials perpendicular to the field lines; in both, the potential is zero at infinity and the field strength is the potential gradient.

Different: gravity only ever attracts, because there is no negative mass, so gravitational potential is never positive. Electric forces can attract or repel. And gravity is extraordinarily weak by comparison: it only dominates on large scales because big objects are almost exactly neutral, so their electric forces cancel while their gravity adds up.

WORKED EXAMPLE

In a hydrogen atom an electron and a proton attract each other electrically and gravitationally. Show that the electric force is about 10^{39} times bigger. ($e = 1.60 \times 10^{-19}$ C, $m_p = 1.67 \times 10^{-27}$ kg, $m_e = 9.11 \times 10^{-31}$ kg)

Show solution

Both forces have r^2 underneath, so it cancels when you divide one by the other:

$$\frac{F_E}{F_G} = \frac{e^2/4\pi\varepsilon_0 r^2}{Gm_p m_e/r^2} = \frac{8.99 \times 10^9 \times (1.60 \times 10^{-19})^2}{6.67 \times 10^{-11} \times 1.67 \times 10^{-27} \times 9.11 \times 10^{-31}} = 2.27 \times 10^{39}$$

This ratio doesn't depend on how far apart they are, because both are inverse-square laws.

Quick checks

QUICK CHECK

The gravitational field strength at the Earth's surface is 9.81 N kg⁻¹. What is it at a height above the surface equal to the Earth's radius?

- 4.91 N kg⁻¹
- 2.45 N kg⁻¹
- 1.09 N kg⁻¹
- 9.81 N kg⁻¹

The distance from the **centre** doubles (from R to $2R$), so g falls by a factor of $2^2 = 4$: $9.81/4 = 2.45\ \text{N kg}^{-1}$. 4.91 forgets the square; 1.09 uses $3R$.

QUICK CHECK

A satellite is moved from a low circular orbit to a higher circular orbit around the Earth. What happens to its speed and its period?

- Speed increases, period increases
- Speed increases, period decreases
- Speed decreases, period increases
- Speed stays the same, period increases

$v = \sqrt{GM/r}$, so a bigger r means a lower speed. It also has further to go, so $T = 2\pi r/v$ increases on both counts ($T^2 \propto r^3$).

QUICK CHECK

Why do astronauts on the International Space Station feel weightless?

- The gravitational field strength at that height is zero.
- They are outside the Earth's atmosphere.
- The Moon's gravity cancels the Earth's gravity.
- They and the station are in free fall with the same acceleration.

g at the ISS is about 8.7 N kg^{-1}, nearly the surface value. Everything on board is falling round the Earth together, so nothing presses on anything else.

QUICK CHECK

At a point in the Earth's field the gravitational potential is −50 MJ kg^{-1}. What is the minimum energy needed to move a 2.0 kg mass from there to infinity?

- 25 MJ
- 50 MJ
- 100 MJ
- −100 MJ

The potential at infinity is zero, so $\Delta V = 0 - (-50) = +50\ \text{MJ kg}^{-1}$. Work done $= m\Delta V = 2.0 \times 50 = 100\ \text{MJ}$. You have to put energy in, so it is positive.

QUICK CHECK

At the neutral point between the Earth and the Moon, which statement is correct?

- Both the field strength and the potential are zero.
- The field strength is zero but the potential is negative.
- The potential is zero but the field strength is not.
- The field strength is at a maximum.

Field strengths are vectors and cancel there. Potentials are scalars and simply add: both are negative, so the total is negative. The potential has a maximum at that point, so its gradient (the field) is zero.

QUICK CHECK

Moon X orbits a planet at four times the orbital radius of moon Y. How many times longer is the period of X than the period of Y?

- 2
- 4
- 8
- 16

$T^2 \propto r^3$, so $T \propto r^{3/2}$. Multiplying r by 4 multiplies T by $4^{3/2} = \sqrt{64} = 8$.